DS 4200 - Machine Learning Systems

Northgate University - School of Data Science - Fall 2026

Instructor: Prof. Ravi Castell (fictional). Meetings: Mon/Wed 4:05 - 5:20 PM, Data Science Building 110.

How deadlines work

Assignments are released on the Monday of the week listed below and are due at the end of the stated number of weeks after release. Weeks are counted from the first week of classes, so an assignment released in Week 2 and due two weeks after release is due at the end of Week 4.

Assignment	Released	Due
Assignment 1: Data pipelines	Week 2	two weeks after release
Assignment 2: Training at scale	Week 3	three weeks after release
Assignment 3: Model serving	Week 6	two weeks after release
Assignment 4: Monitoring and drift	Week 9	three weeks after release

Course project: teams of two submit a one-page project proposal by the end of Week 7 and a final report by the end of Week 15. There is a single in-class exam in Week 11.

Grading

Component	Weight
Assignments 1-4 (10% each)	40%
Project proposal	5%
Project final report	25%
Exam	30%

Topics by week

Week 1 introduction; Week 2 data pipelines; Week 3 distributed training; Week 4 hyperparameter search; Week 5 evaluation; Week 6 serving; Week 7 batching and caching; Week 8 feature stores; Week 9 monitoring; Week 10 drift; Week 11 exam; Weeks 12-15 project work and guest lectures.

Academic Integrity

All work submitted for credit must be your own. Collaboration on concepts is encouraged, but written answers and code must be produced independently unless an assignment explicitly allows teams. Suspected violations are referred to the Office of Student Conduct under the Northgate Honor Code.

Accessibility

Students who need accommodations should register with the Northgate Accessibility Resource Center and share their letter with the instructor as early as possible in the term.

Late Work

Unless stated otherwise, late submissions lose 10% per day and are not accepted after three days. Extensions for documented emergencies are granted on request.

Course Evaluations

You will be invited to complete an anonymous course evaluation near the end of the term. Feedback is read carefully and used to improve the course.